Case Analysis Using JMP: Sleep and Gestation in Mammals

Executive Summary

The goal of this case study is to analyze whether gestation length predicts REM sleep, Non-REM sleep and total sleep using a data from 79 mammalian species. It is hypothesized that gestation duration and sleep patterns will have a negative relationship. This will be explored using regression analysis and K-means clustering in JMP.

Regression analysis indicate that gestation length is negatively associated with total sleep, NREM sleep, and REM sleep, though to a progressively less degree for each. Clustering analysis of the sleep variables identifies 2 clusters of species, which had notably different gestational lengths and variation.

These findings suggest that gestation length plays a statistically significant moderate role in influencing mammalian sleep patterns. Understanding these relationships may inform future comparative studies of mammalian sleep patterns and gestation, especially in uncovering other similarities and differences between the two clusters that were found.

Introduction and Overview

Much research has been done on mammalian sleep patterns and what factors cause variation among species. Alain A. Gonfalone from the European Space Agency hypothesized that the length of the gestation period is one of the key factors that determine the amount of REM and Non-REM sleep mammals experience (JMP Statistical Discovery, 2023). Gonfalone's hypothesis will be explored in this analysis using JMP. The null hypothesis is that there is no relationship between gestation length and sleep patterns.

The JMP dataset for this topic consists of 79 mammalian species with measurements of gestation time, total daily sleep, Non-REM sleep, and REM sleep (see Table 1). In total, there are 79 instances in the dataset, one for each species. Each value in the data columns below is the average for the species. Below are the first 15 instances of the dataset.

Table 1: Summary of Sleep and Gestation Variables	
Mammal	Species name
Gestation	Time of average gestation (days)
TST	Average total daily sleep (hours)
NREM	Average amount Non-REM sleep (hours)
REM	Average amount of REM sleep (hours)

	Mammal	Gestation	TST	NREM	REM
1	African giant pouched rat	34.4	8.3	6.3	2
2	Arctic ground squirrel	25	16.08	12.7	3.38
3	Barbary macaque	164.6	11.75	10.68	1.07
4	Belding's ground squirrel	26.8	15.84	12.83	3.01
5	Big brown bat	47.5	19.7	15.8	3.9
6	Black lemur	132.5	9.65	8.81	0.84
7	Blind mole rat	•	12.44	10.54	1.9
8	Bonnet macaque	76.8	9.17	8.12	1.05
9	Brandt's hedgehog	37	10.32	7.5	2.82
10	Brushtail possum	18	13.68	12	1.68
11	Domestic cat	63.9	13.23	10.01	3.22
12	Chimpanzee	223.8	10.8	9.18	1.62
13	Chinchilla	110.8	12.52	10.97	1.55
14	Collared lemming	18	15.19	12	3.19
15	Common genet	77.8	6.1	4.8	1.3

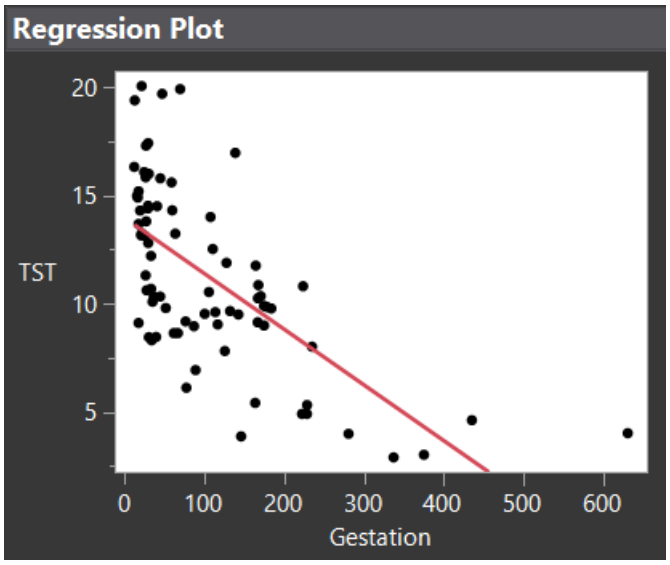
The objective of this analysis is to evaluate whether gestation length is significantly related to sleep duration (TST, NREM, and REM) using regression analysis, and to identify natural clusters of species based on their sleep patterns using K-means clustering. This will reveal patterns between species related to gestation length and sleep patterns.

Methods and Analysis

I. Linear Regression

Simple linear regression is ideal to test if there is a relationship between gestation length, total sleep, NREM sleep, and REM sleep across species. The strength and direction of the relationship will also be determined.

a. Average total daily sleep (TST) and gestation length (Gestation):



Summary of Fit

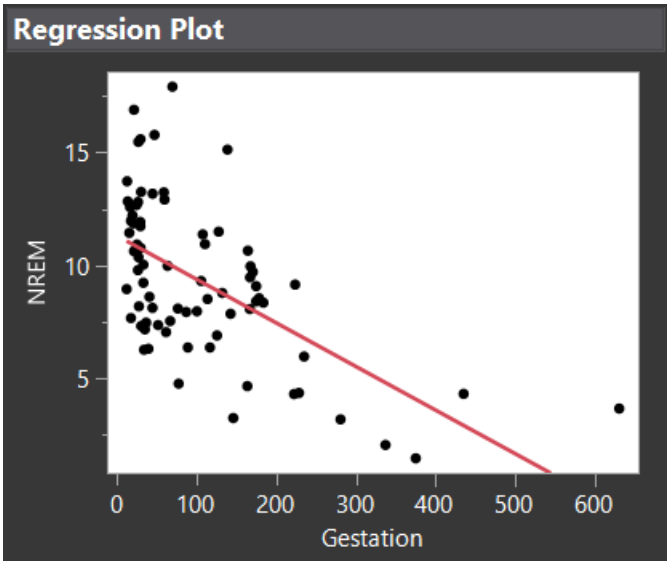
RSquare	0.43102
RSquare Adj	0.423533
Root Mean Square Error	3.218363
Mean of Response	11.2859
Observations (or Sum Wgts)	78

Parameter Estimates

Term	Estimate	Std Error	t Ratio	Prob>|t|
Intercept	13.972783	0.508123	27.50	<.0001*
Gestation	-0.025811	0.003402	-7.59	<.0001*

Above is the scatterplot and regression for average total daily sleep and gestation length. The summary of fit shows R^2=0.431, which indicates 43% of the variation in TST is explained by gestation length. The average TST for all species is 11.29 hours per day. The parameter estimates above show the direction and magnitude of the relationship, with a slope of -0.0258. This means that, on average, each additional day of gestation is associated with a predicted decrease in TST of 0.0258 hours, or roughly 1.5 minutes. The p value < 0.0001 shows that this negative relationship is statistically significant and unlikely to be due to chance.

b. Average Non-REM (NREM) and gestational length (Gestation)



Summary of Fit

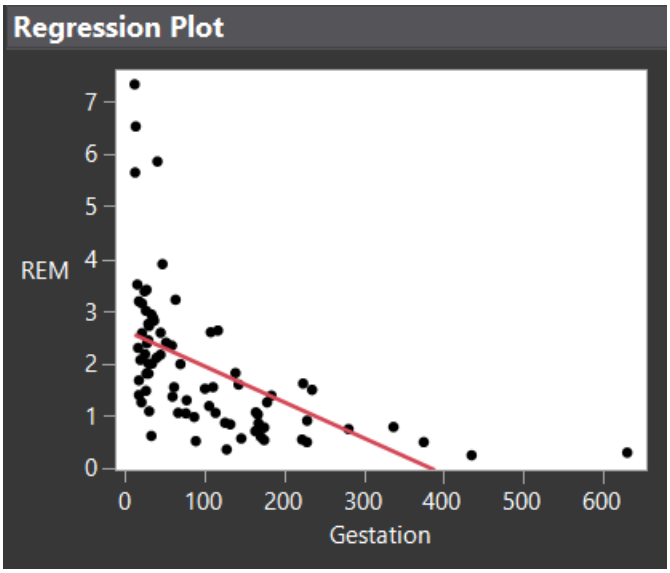
RSquare	0.357792
RSquare Adj	0.349342
Root Mean Square Error	2.808057
Mean of Response	9.342564
Observations (or Sum Wgts)	78

Parameter Estimates

Term	Estimate	Std Error	t Ratio	Prob>|t|
Intercept	11.353034	0.443343	25.61	<.0001*
Gestation	-0.019313	0.002968	-6.51	<.0001*

Above is the scatterplot and regression for average Non-REM sleep and gestation length. The summary of fit shows R^2 = 0.358, meaning gestation length explains about 36% of the variance in NREM sleep length. The average hours of NREM sleep among all species is 9.34 hours per day. The slope is -0.0193, showing the relationship between gestation length and NREM sleep is also negative, though it is not as strongly related as gestation length and TST. This makes sense because TST is made up of mostly NREM sleep and some REM sleep, so any relationship between TST and gestation will be reflected in NREM, albeit to a smaller degree. Statistical significance for the relationship is demonstrated again, with $P < 0.0001$.

c. Average REM and gestational length (Gestation)



Summary of Fit	
RSquare	0.287274
RSquare Adj	0.277896
Root Mean Square Error	1.17482
Mean of Response	1.924026
Observations (or Sum Wgts)	78

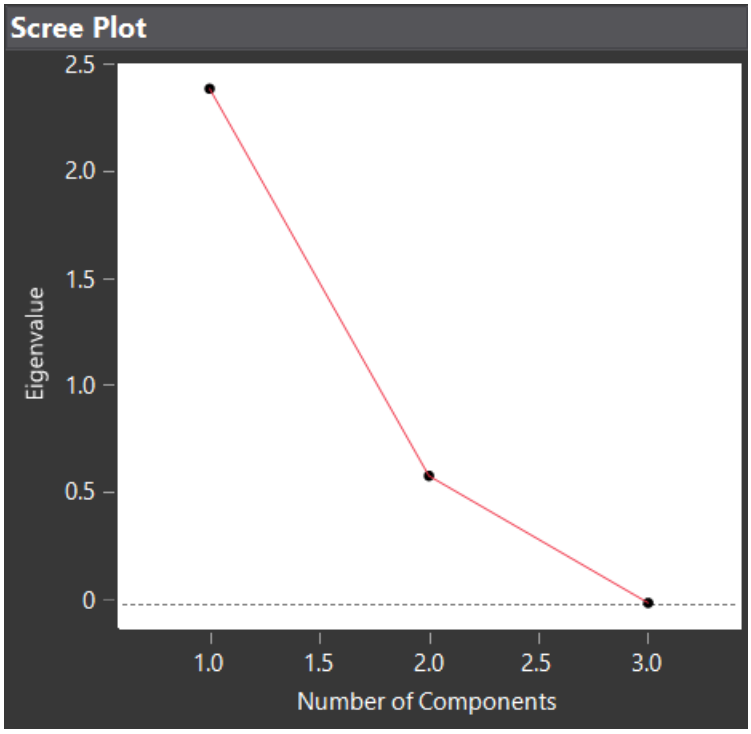
Parameter Estimates

Term	Estimate	Std Error	t Ratio	Prob>|t|
Intercept	2.6394639	0.185484	14.23	<.0001*
Gestation	-0.006873	0.001242	-5.53	<.0001*

Lastly, there is the regression for average REM and gestational length. Like the relationships shown for NREM and TST, the plot above shows a negative relationship between average REM and Gestation, though it is weaker than the other sleep stages. $R^2 = 0.287$, indicating that gestation explains about 29% of the variation in REM sleep. The average REM among all species observed is 1.92 hours per day. The slope is much smaller as well, -0.0069, meaning that, on average, each additional day of gestation predicts about 0.4 minutes less REM sleep per day. This relationship is also statistically significant ($P < 0.001$).

II. K-means Clustering

K-means clustering is used in this analysis because it identifies natural groups of mammals based on similar sleep patterns. This unsupervised ML method is useful because it can show patterns not captured by regression by itself, revealing other aspects of biology or reproductive strategies that can be later studied. This dataset is limited to purely sleep stage duration and gestation, but species clustering can reveal other variables that could be further analyzed.

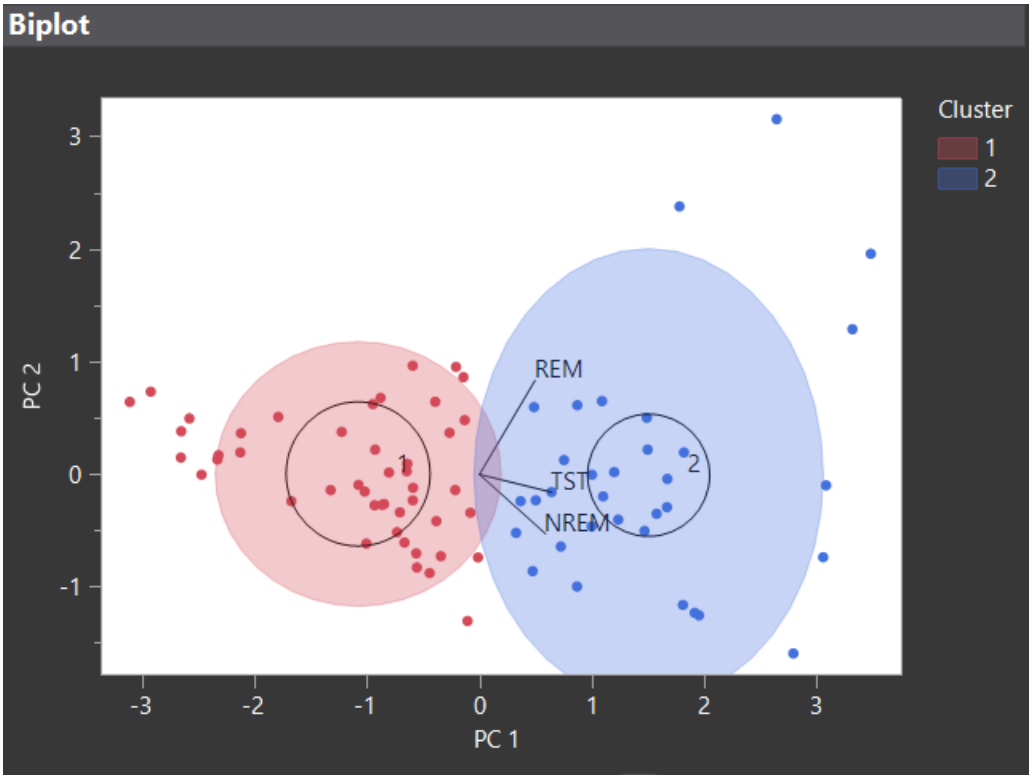


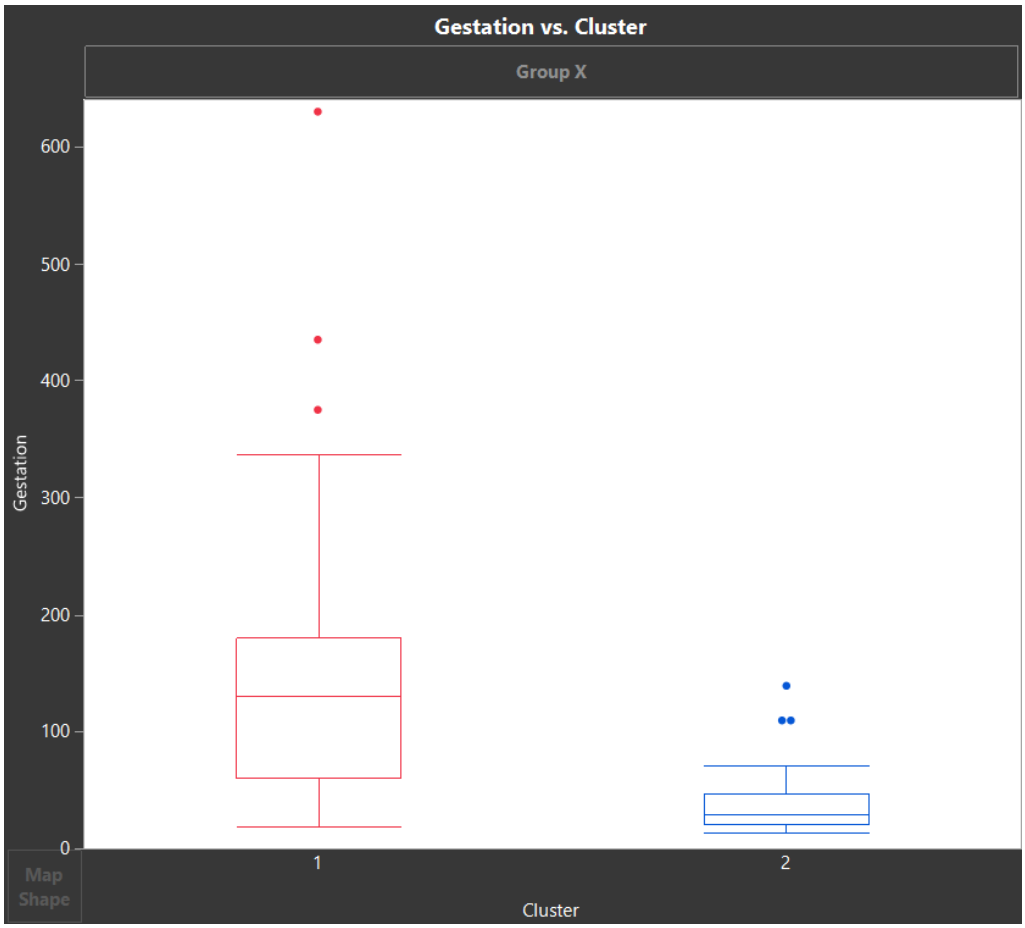
Eigenvalues

Number	Eigenvalue	Percent	20 40 60 80	Cum Percent
1	2.404856	80.162		80.162
2	0.594685	19.823		99.985
3	0.000459	0.015		100.000

Above are the eigenvalues and the scree plot for the three sleep types: TST, NREM, and REM. 99.085% of the variance in the three sleep variables is explained by the first two eigenvalues, as shown in the eigenvalues table. Likewise, using the elbow method in the scree plot, the curve flattens after the second point. These data show the optimal number of clusters for K-means is k=2.

Below is the Biplot of the K-means clusters, showing two clusters: cluster one is more condensed, cluster two has more variation and outliers are more widely distributed. These clusters were formed based on sleep patterns only (TST, NREM, and REM). To compare them to gestation, the clusters were saved and graphed against gestation in a box plot. Though there is lower variation in point distribution for cluster one, gestation variation for cluster one is much higher. Gestation values are generally higher, averaging at 150 days, and there is a wider interquartile range, 60-180 days. Cluster two has much less gestational variation, even though its cluster is more widely spread than cluster one. Gestation values are much lower and tightly grouped, with an average gestation of 29 days and an interquartile range of 20-47 days.


Biplot
Cluster
1
2
PC 2
PC 1
REM
TST
NREM
1
2
3
2
1
0
-1
-3
-2
-1
0
1
2
3


Gestation vs. Cluster
Group X
Gestation
600
500
400
300
200
100
0
1
2
Cluster
Map Shape

Discussion and Conclusions

From the linear regressions and K-means clustering, a relationship between gestation length and sleep patterns emerges, though it only explains sleep variation moderately. Average TST consists of Non-REM and REM sleep. But Non-REM makes up most of the total sleep, so TST and NREM regressions look very similar, and the REM regression shows a weaker, but still negative relationship to gestation. This pattern aligns with the R^2 values: TST R^2=0.43, NREM R^2=0.36, and REM R^2 = 0.29. Gestation is most related to total sleep, but that's because total sleep is made of mostly Non-REM sleep. REM sleep appears to be less dependent on gestation, and its variance is due to other factors.

The negative slopes from the linear regressions clearly show that, as average gestation length increases, TST (and, more specifically, NREM duration) is predicted to decrease. This matches other studies and validates Gonfalone's hypothesis, though it is apparent that other factors also explain the variation in the sleep patterns of the 79 mammals. Further analysis could include additional factors, such as body size, diet, and metabolic rate. Exploring these factors in conjunction with gestation could help us understand what explains the remaining 57% of the variation in total average sleep duration. Additionally, other analyses, such as PCA, more detailed clustering, and a Prediction Profiler, could show how these variables interact to determine sleep patterns.

Graphing the clusters against gestation length shows that cluster one consists of species with similar sleep patterns but diverse gestation lengths. Cluster two, however, shows a group of species that have a wider range of sleeping patterns but very similar gestation lengths. Cluster two has a lower gestational average length (29 days) than cluster one (150 days). The linear regression determined that as gestation increases, total average sleep decreases, so we would expect that animals in cluster one would sleep less than animals in cluster two. Continuing research could explore what causes the variation in clustering between the two groups. Gestation doesn't explain enough of the variation to determine a biological interpretation for the cluster spreads. There may be other unifying variables that cause cluster one to have converged on such similar sleeping patterns. Likewise, there may be more diversity in the ecosystem, physiology, or niche within cluster two that is driving the variation in sleeping patterns.

Researching other potential variables that explain these trends would reveal insights that this analysis's limitations leave unexplained. Comparing only sleep length data and gestation limits the biological interpretations that can be made. Other aspects of sleep quality could provide more insight into gestation, and other biological variables could provide a better explanation for sleep length.